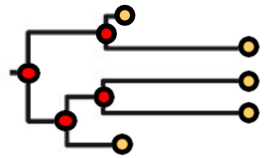
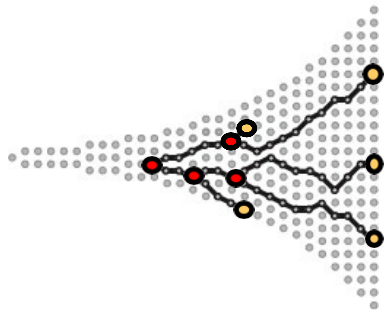
Infectious Disease 'Omics

Introduction to phylodynamics and populations

Lecture Overview

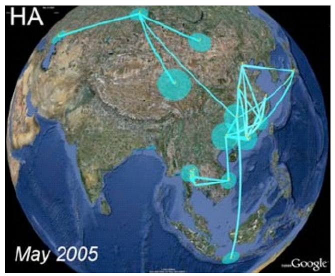


1) Introduction



2) Molecular Dating

3) Population Dynamics

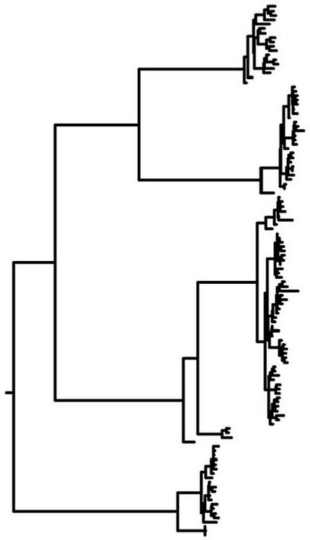


4) Phylogeography

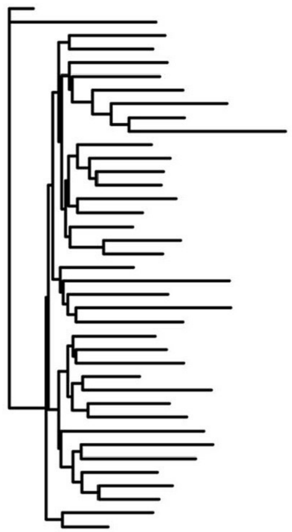
Introduction

Unifying epidemiology and evolution

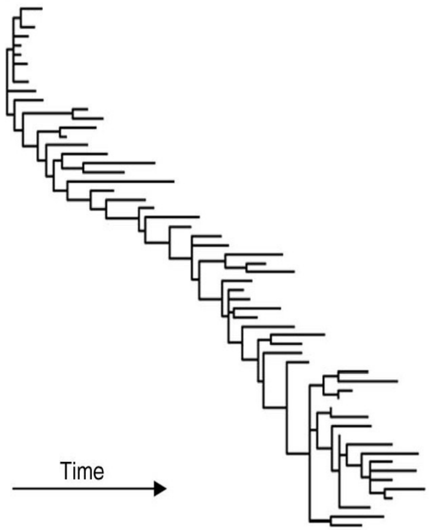
Measles virus
population phylogeny



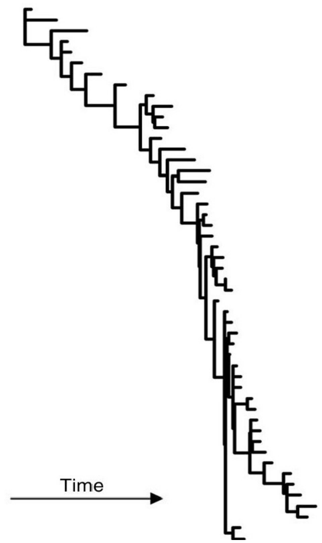
Dengue virus
population phylogeny



HIV
population phylogeny



HIV
within host phylogeny

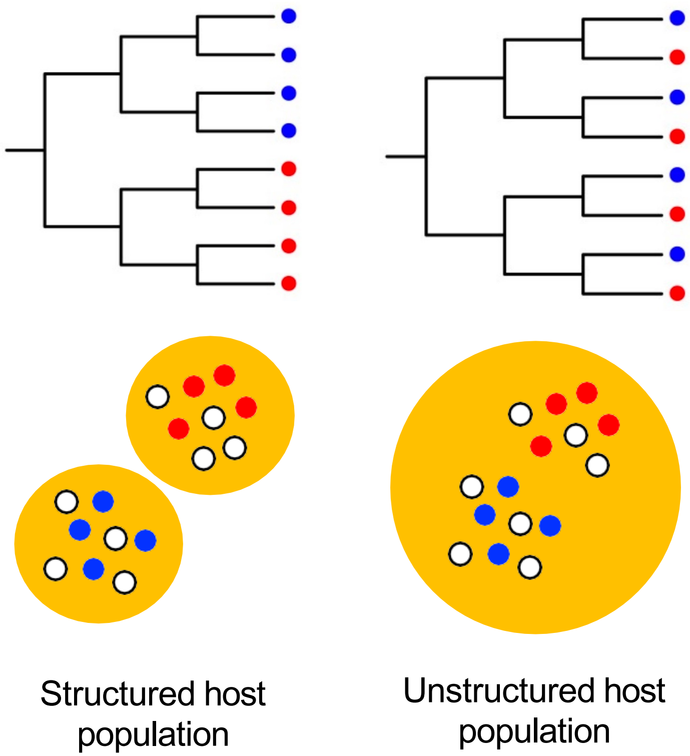


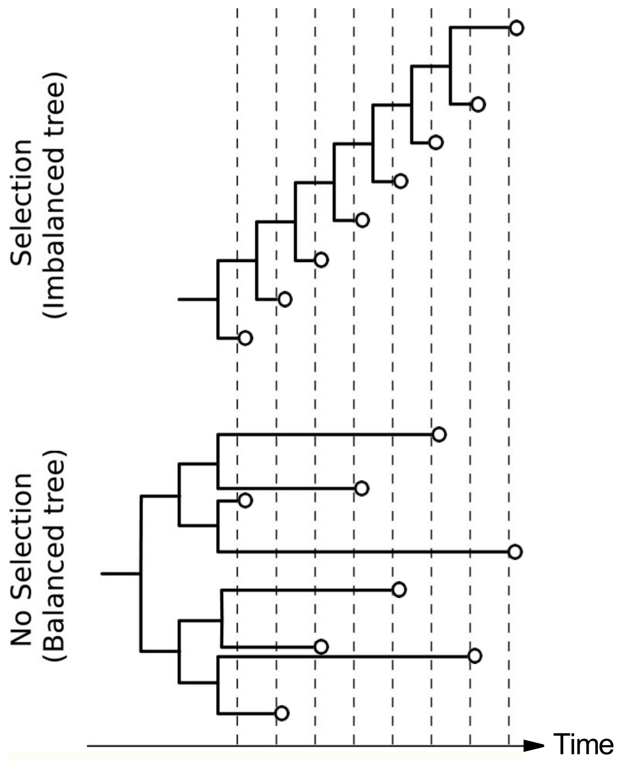
Influenza A virus
population phylogeny

Grenfell et al. 2004. Science

Unifying epidemiology and evolution

- **Phylodynamics:** the study of how epidemiological, immunological and evolutionary processes shape viral phylogenies





Volz et al. 2013. PLoS Comput Biol

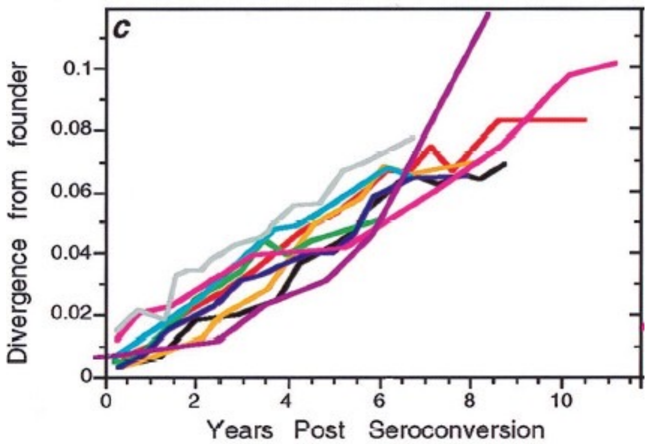
Molecular Dating

The Molecular Clock Hypothesis

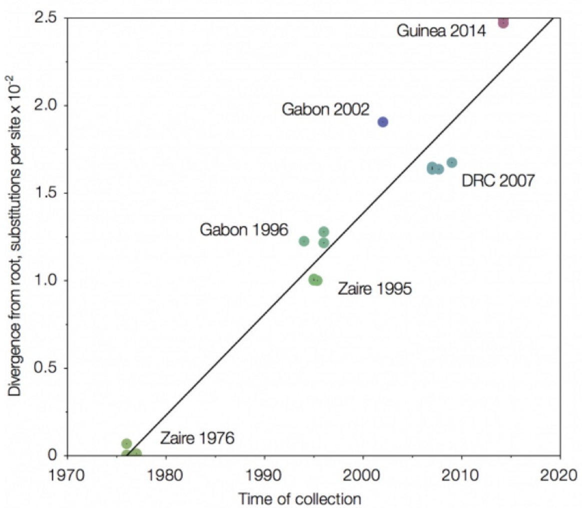
- For a given organism, gene or strain, mutations are fixed at a (nearly) **constant pace**
- Molecular Clock models define a relationship between genetic distance and time:

- ***Strict clock:*** all lineages evolve at the **same rate**

- ***Relaxed clock***: Allows evolutionary rates to **change** over time and along lineages

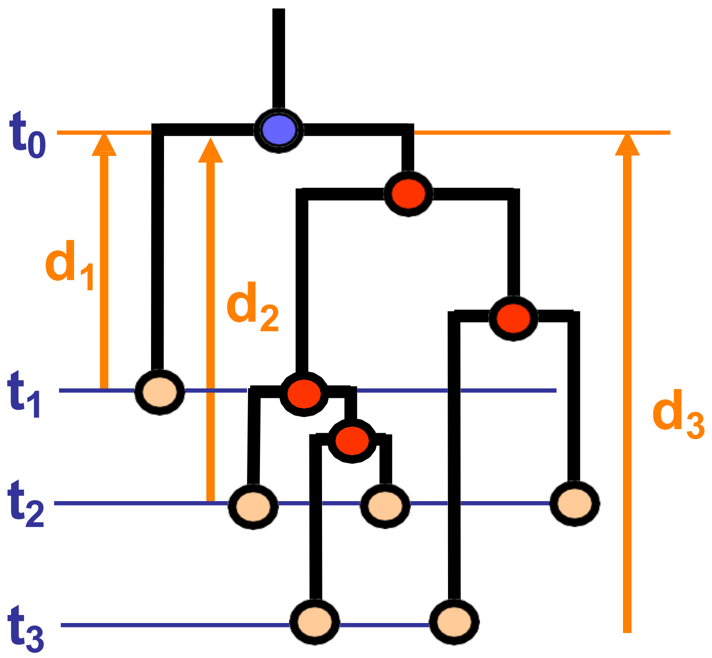


Ebolavirus between host evolution

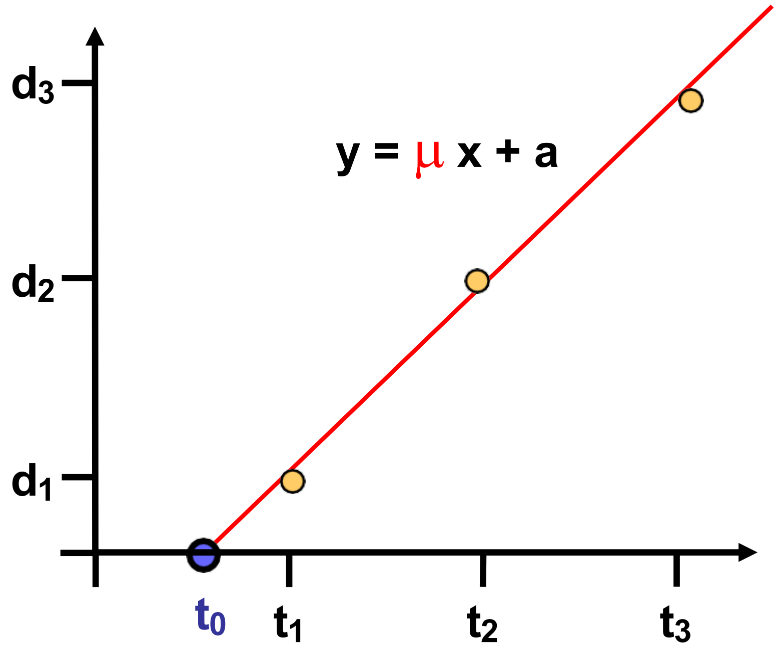


Shankarappa. 1999. J Virol; Dudas & Rambaut. 2014. PLoS Curr.

Estimating Substitution Rates



Phylogeny



Root-to-Tip linear regression

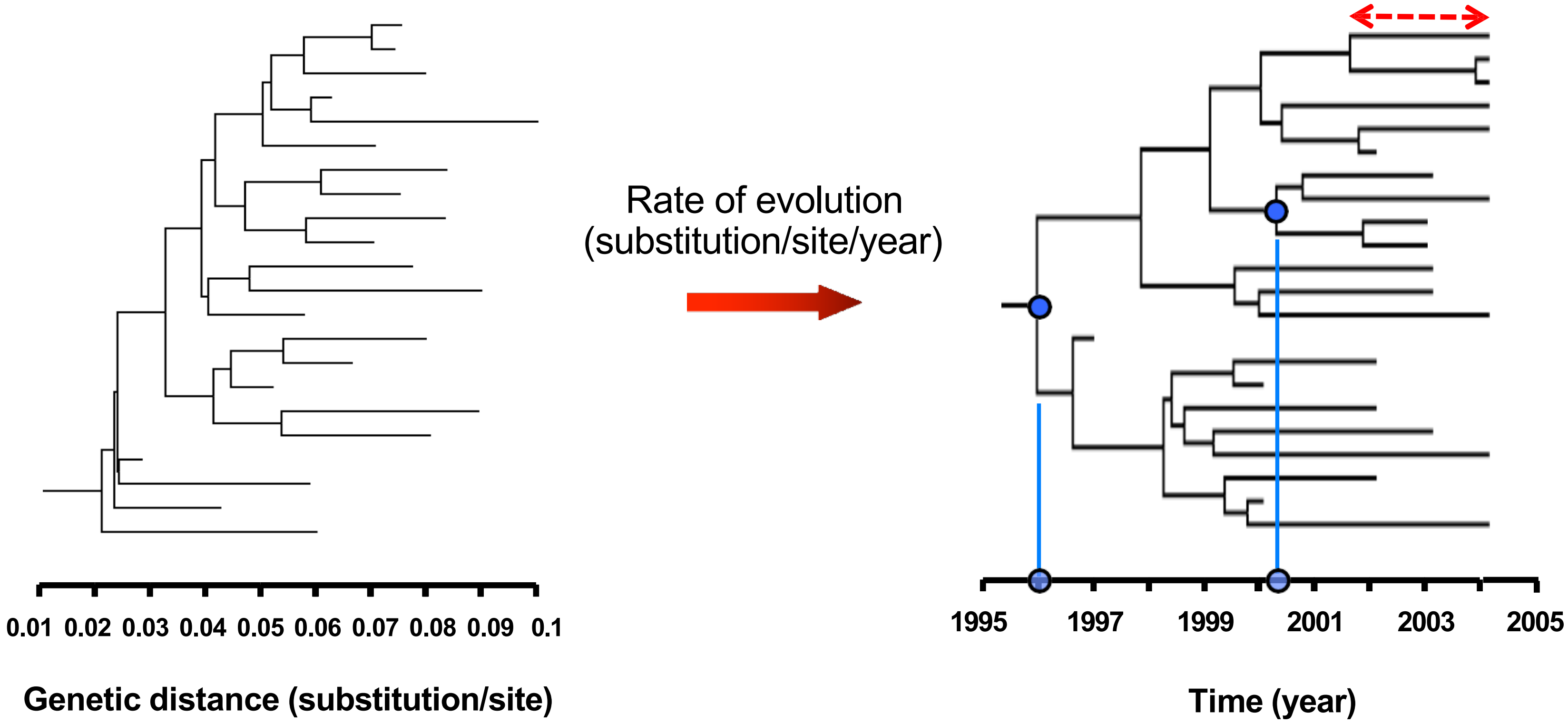
d_i : genetic distance between sequence and root

t_i : Sampling time of sequence *i*

Evolution rate: $\mu = \Delta d / \Delta t$

Time of origin: t_0

From Genetic Diversity to Time



- Date the **origin** of an epidemic (root) or outbreak (internal node)
- Estimate transmission **intervals** (internal branch lengths)

Statistical Inference of Evolution Rates

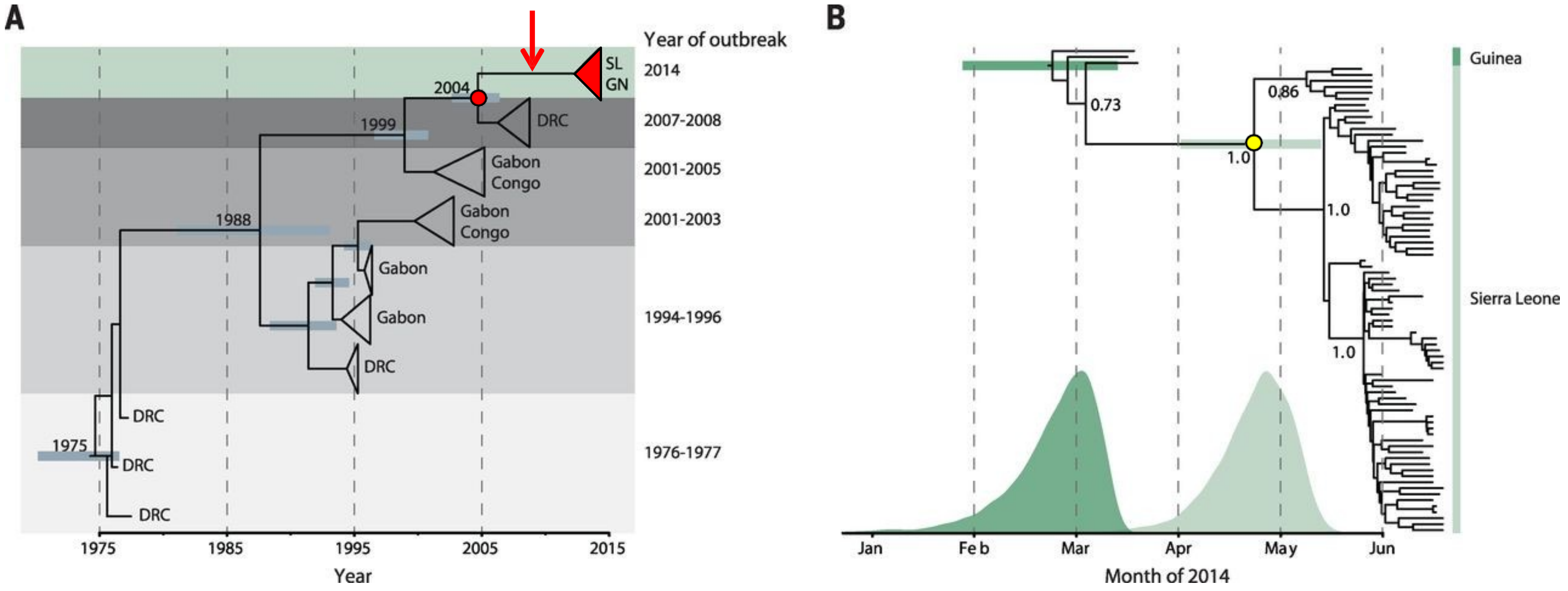
- Root-to-Tip linear regressions rely on two **unrealistic** assumptions:
  - **Constant** evolution rate over time
  - **Independence** of evolutionary histories (violated in internal branches)

 Incorporation of phylogenetic structure & **statistical hypothesis testing**:

1) Each tip of the tree has a known time
2) Internal nodes are given arbitrary starting times consistent with their order in the tree
3) Vector of internal nodes times (t_o, t_1, … t_n) and **substitution rate** are optimized until the values providing the **best fit** are found

 Maximum likelihood or **Bayesian** statistical framework

The Guinea 2014 Ebolavirus Outbreak

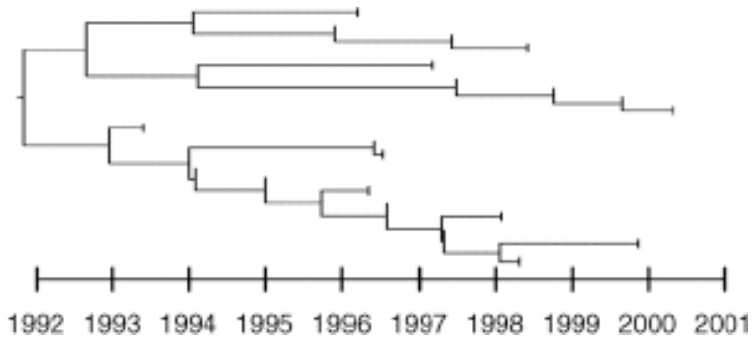


- **Single introduction** from the natural reservoir (→ in **A**)
- Divergence from Central African variant late 2004 (● in **A**), crossed from Guinea to Sierra Leone in **May 2014** (○ in **B**)
- No evidence of further **zoonotic** infections

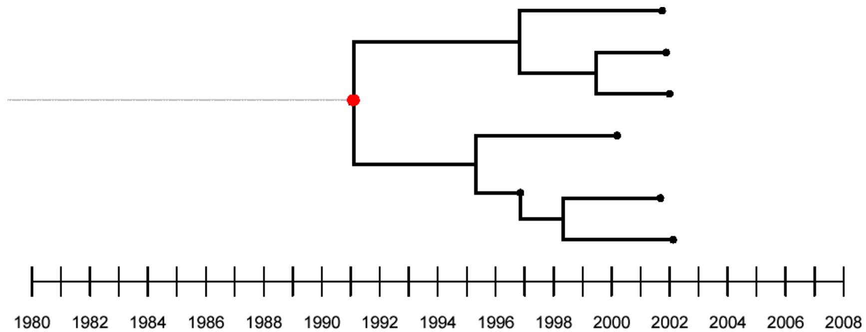
Gire *et al. 2014. Science*

HIV Sexual Transmission in the UK

Sex between men (Subtype B)



Heterosexual contact (Non-B Subtypes)



Subtype	n Patients	Frequency of linkage (> 10 individuals)	Median transmission interval*	Frequency of transmission within 6 months post-infection
B	2 126	25%	14 months	25%
Non-B	11 071	5%	27 months	2%

* Median internode interval within a cluster

Lewis et al. 2008. Plos Med; Hughes et al. 2009. PLoS Path

Population Dynamics

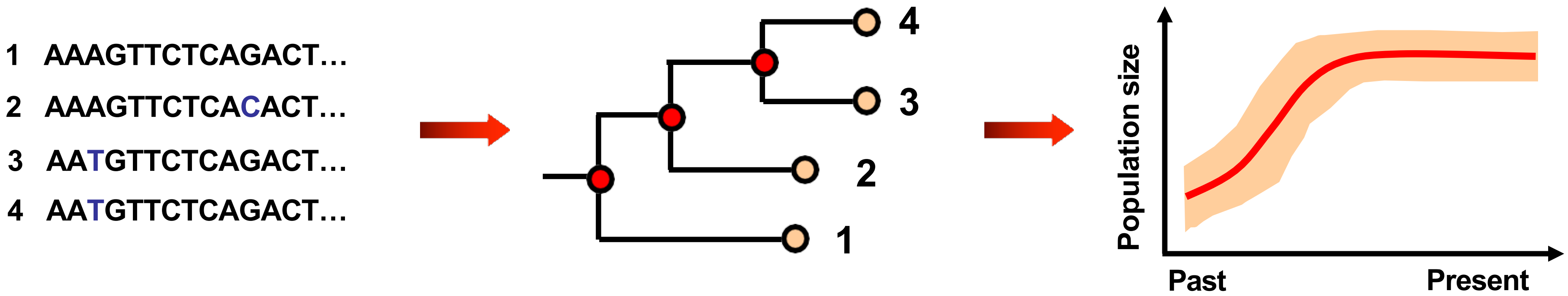
Phylogenies & Population Dynamics

• Inference of **demographic characteristics** from phylogenies:

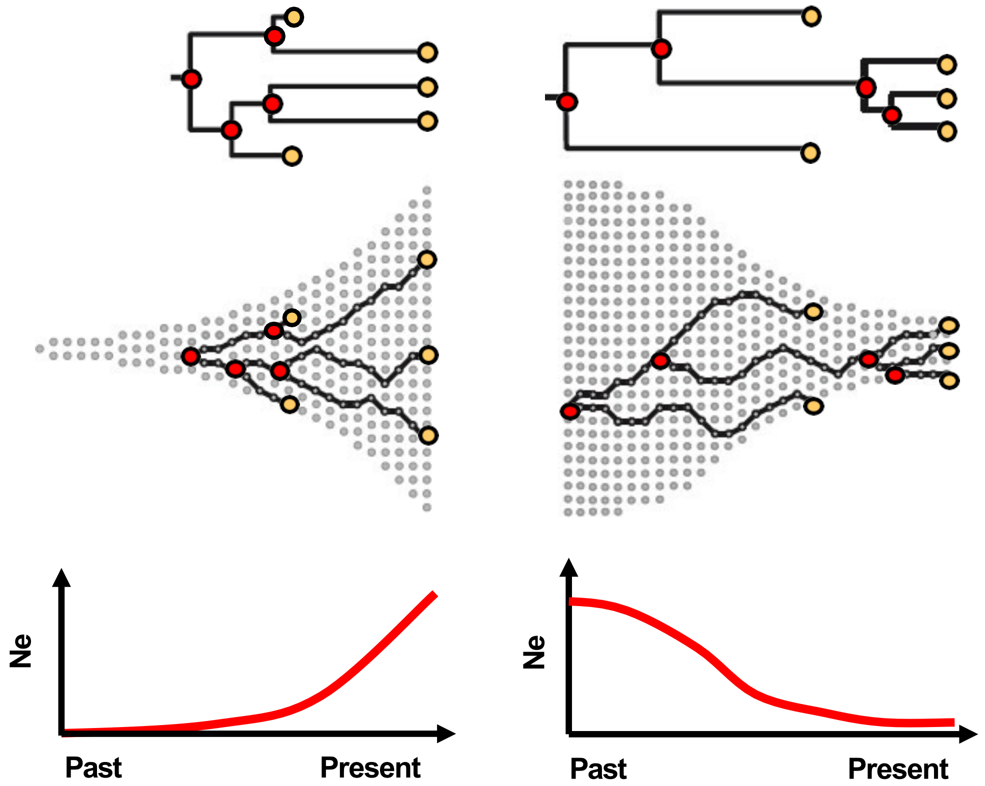
- **Effective population Ne** (fraction of the total population contributing to the next generation)

- **Growth rate** of a population

- **Time of origin** of an epidemic



The Coalescent Theory



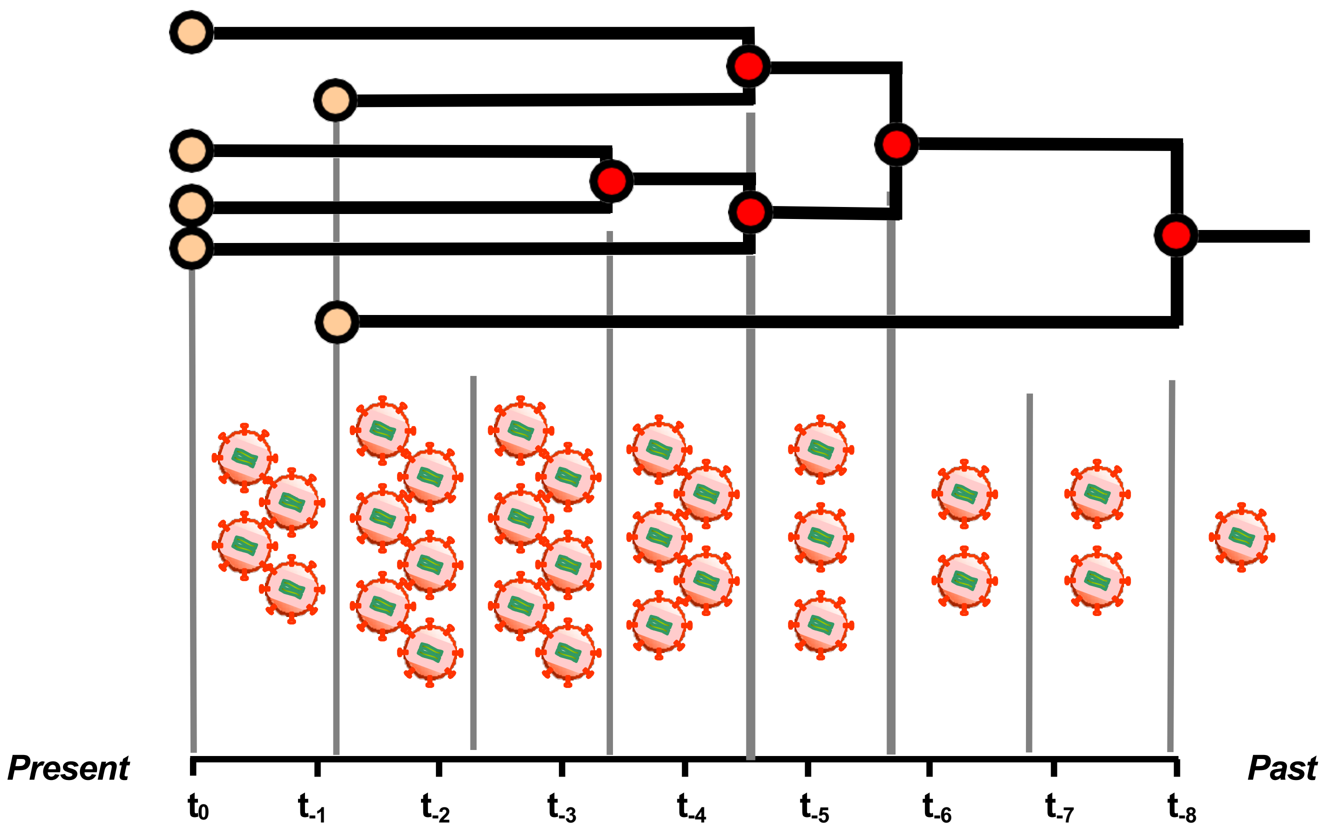
Relationship between genetic diversity and population size:

The rate at which lineages '**coalesce**' depends on **population size** and **population structure**

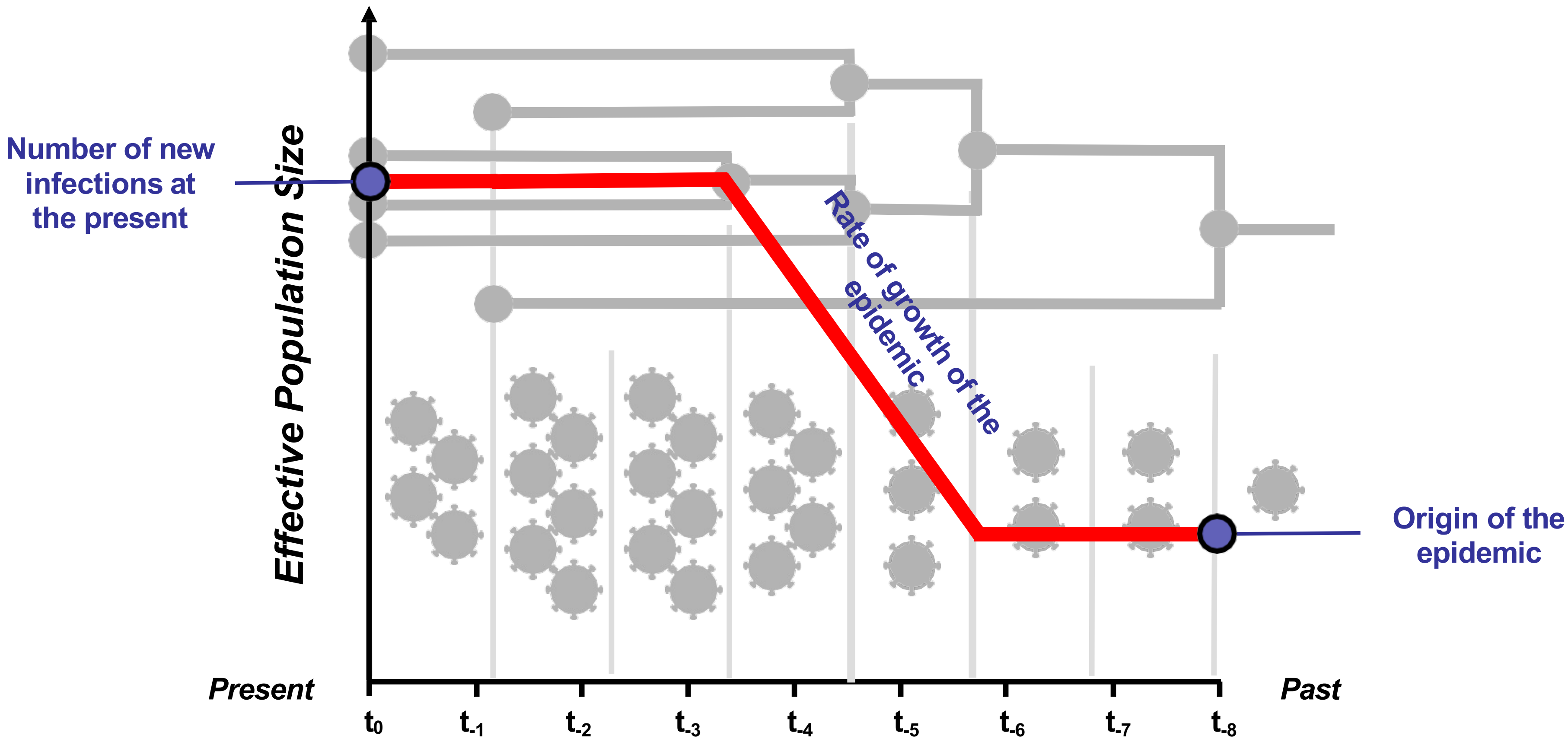
Skyline plot: Effective population time over time

Lemey et al. 2007. AIDS Rev

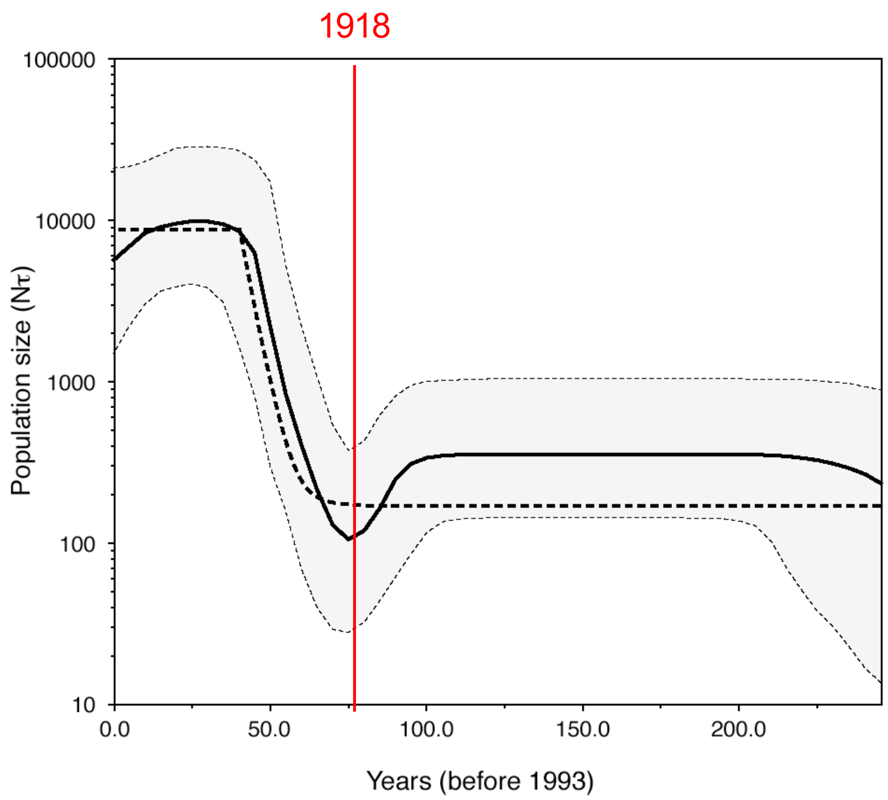
The Coalescent Theory II



The Coalescent Theory II



Hepatitis C virus in Egypt



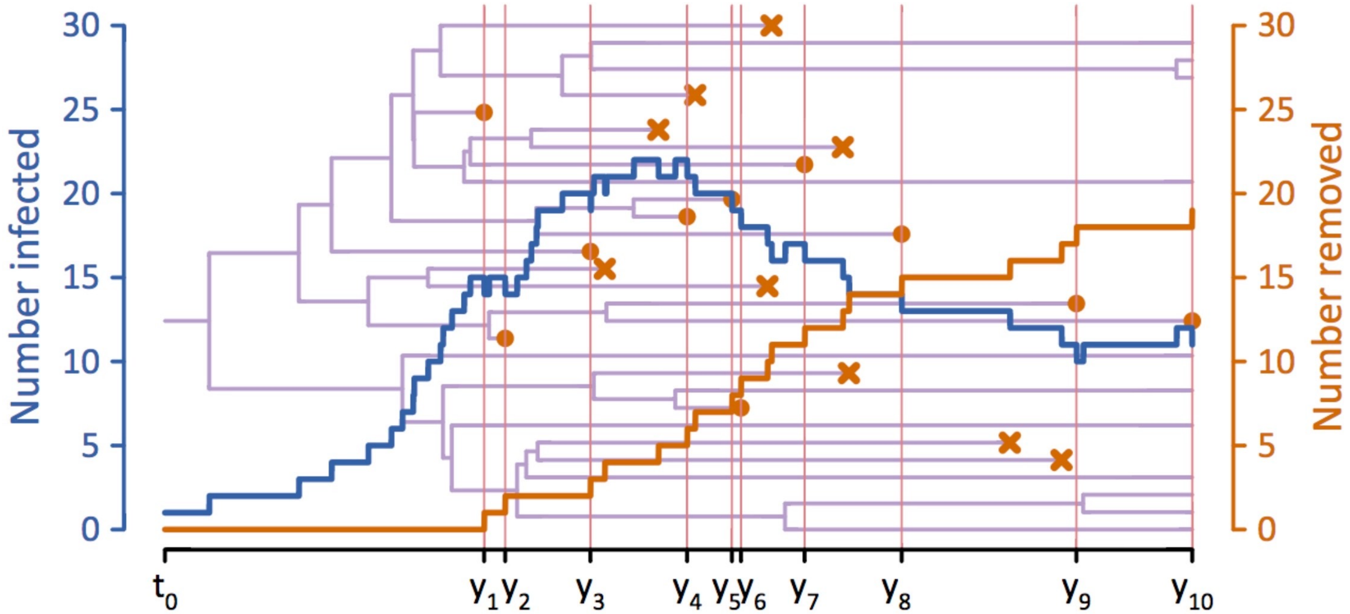
- HCV **demographic history** in Egypt estimated from 63 E1 gene sequences

- Ne dramatically **increased** in the mid 1900s

- Probably caused by **contamination** of injectable antischitosomiasis treatment

Pybus et al. 2003. Mol Biol Evol; Drummond et al. 2005. Mol Biol Evol

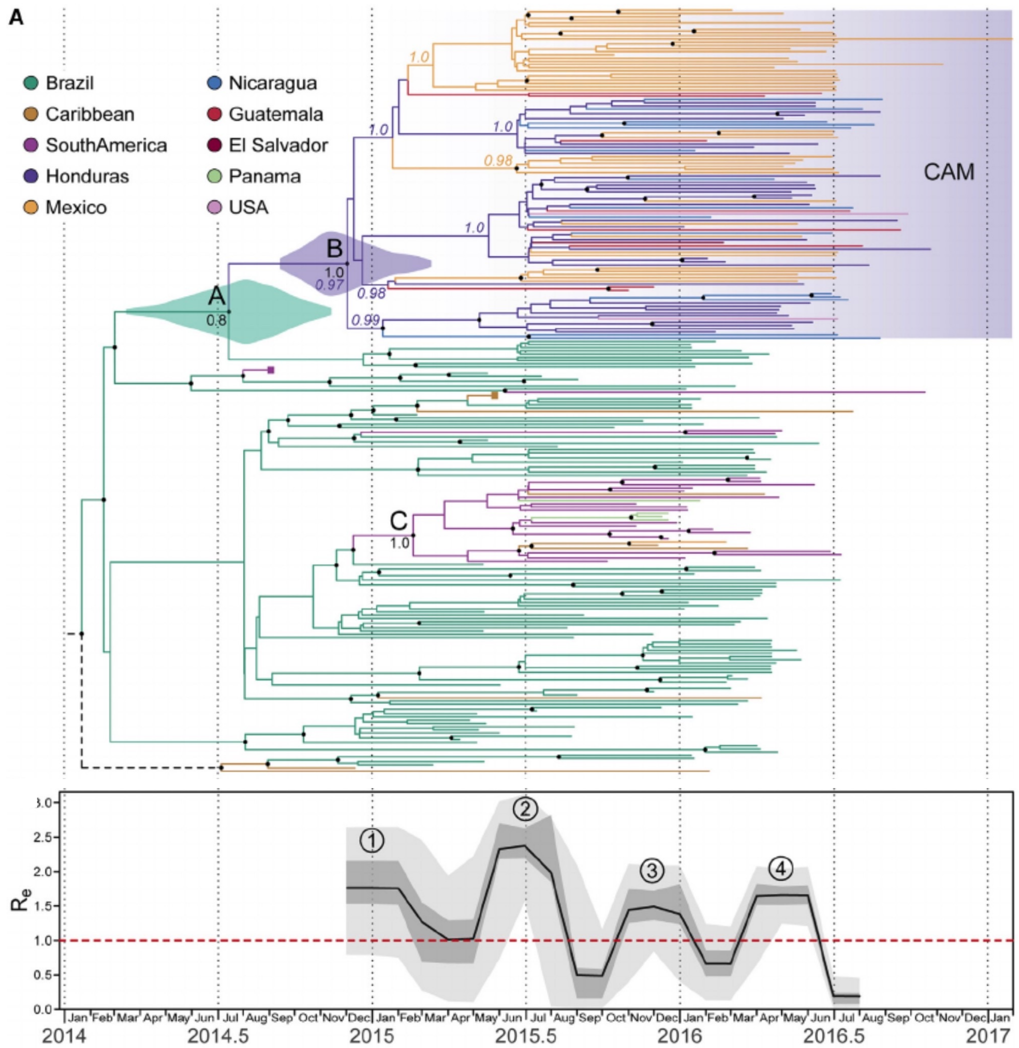
Estimating epidemiological parameters



Number of infectious and removed (i.e., non-infectious) individuals of an epidemic estimated from a viral phylogeny

du Plessis & Stadler. 2015. Trends in Microbiology

Estimating Zika Re in Central America & Mexico



Zika **effective reproductive number** (Re) through time, estimated using a birth-death skyline approach

(1) ZIKV spread from Honduras to other CA countries
(2) Rapid radiation of ZIKV lineages
(3) Period prior to rapid increase in reported ZIKV cases
(4) Observed epidemic (April–July 2016)

Thézé et al. 2018. Cell Host & Microbe

5 – Phylogeography

Reconstructing Geographic Dispersal

• Statistical methods to reconstruct the history of **pathogen spread**:

Geographic location of the pathogen estimated on each branch of a phylogeny

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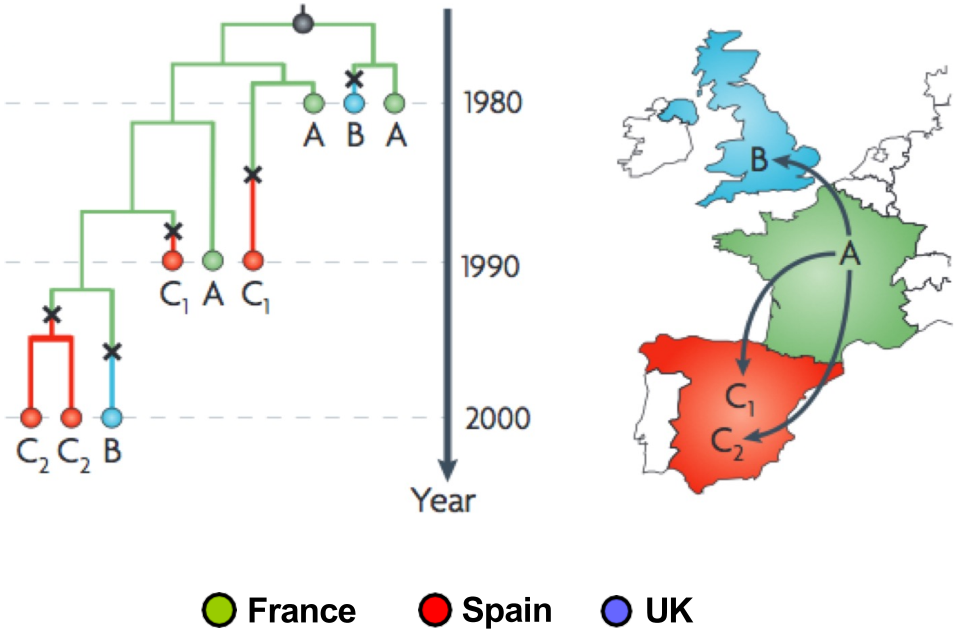


Pybus & Rambaut. 2009. Nat rev Genet

Reconstructing Geographic Dispersal

- Parsimony dismisses phylogenetic **uncertainty** and **unequal** state exchange probability

- **Bayesian MCMC** approach:

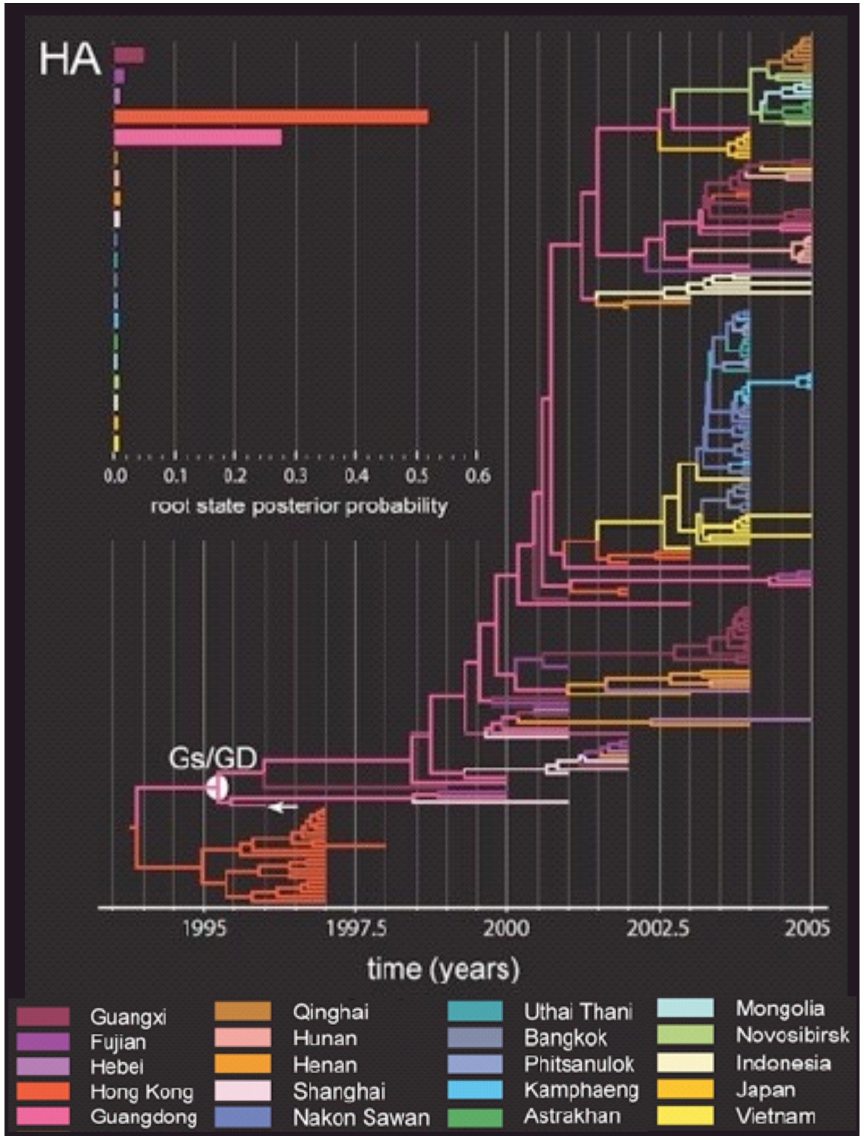
  - **Ancestral reconstruction** of spatial states along a dated phylogeny:

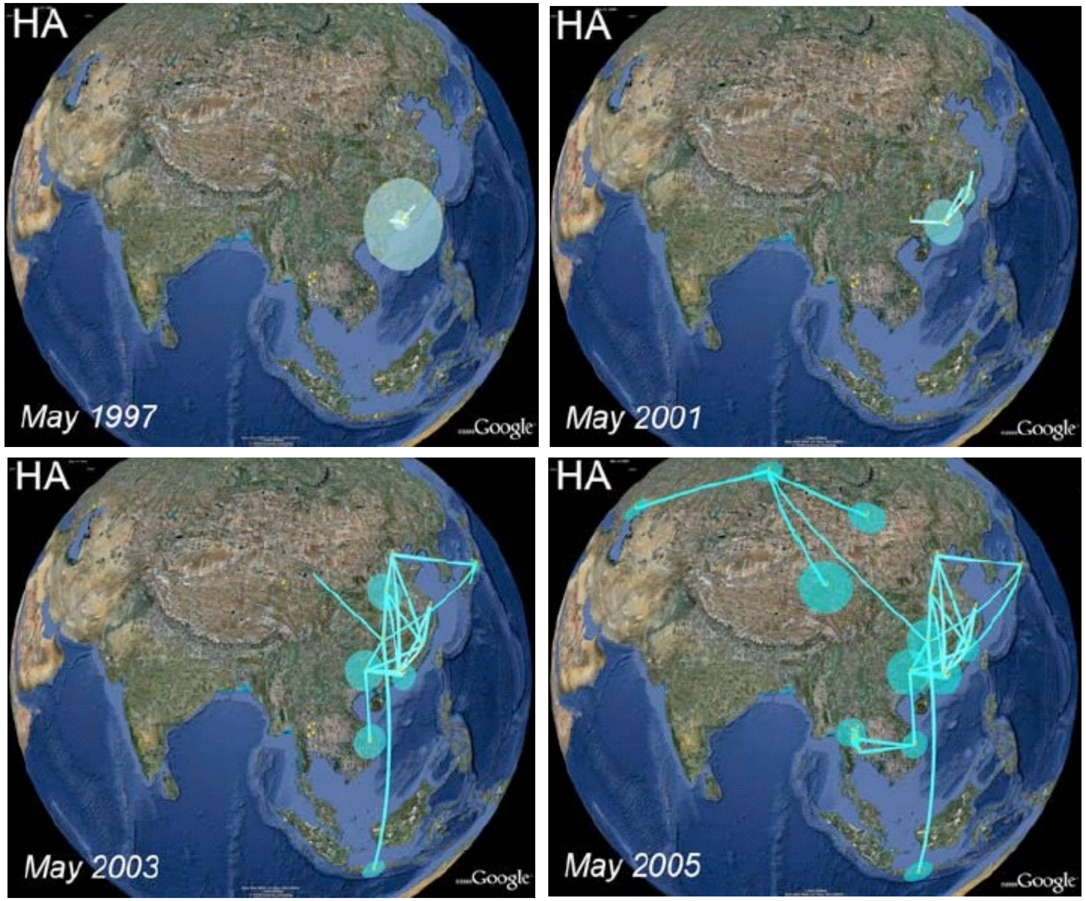
    Spatial mapping in natural time scales

  - Calculate the **probability** of each ancestral state given the observed data

  - Calibrated under strict / relaxed **molecular clock** + **demographic** models

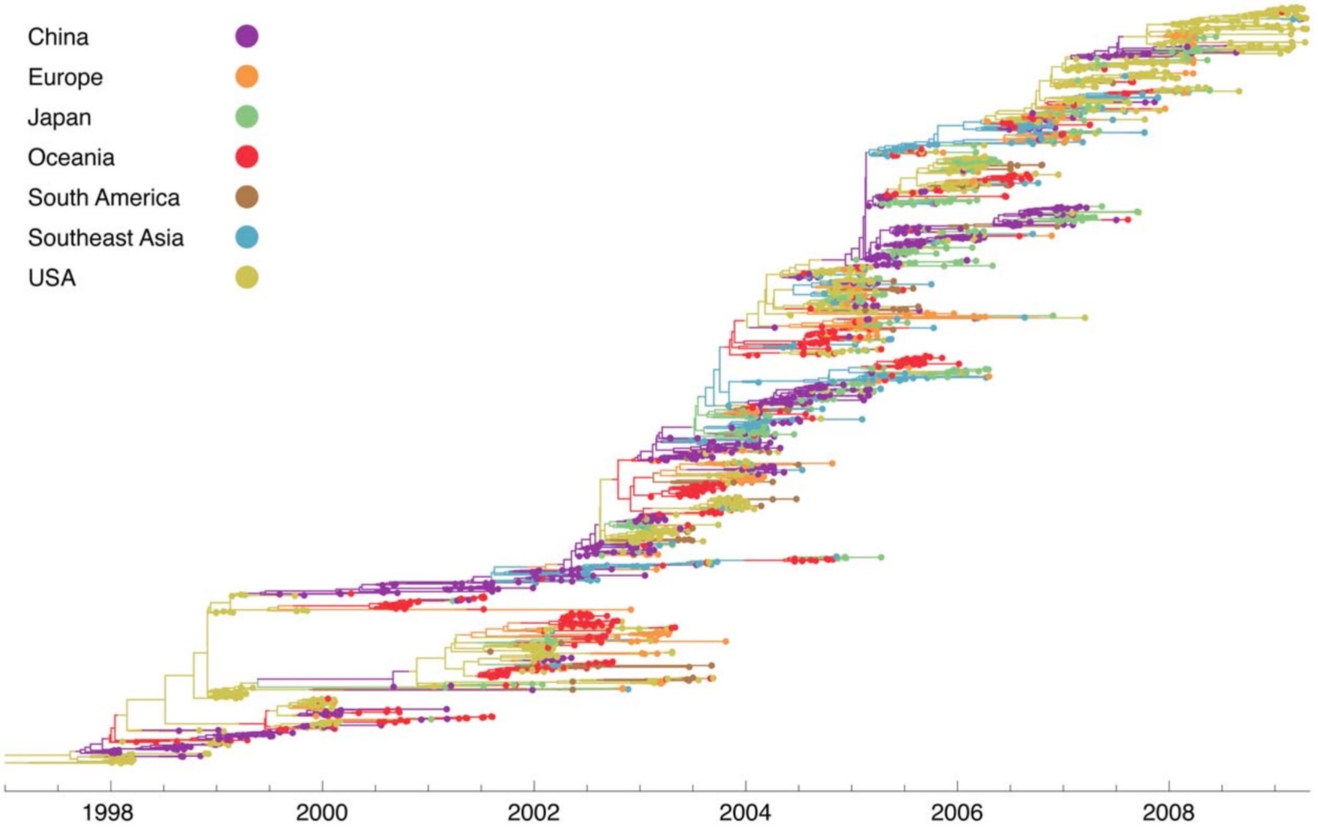
Influenza A H5N1 Dispersal in South East Asia

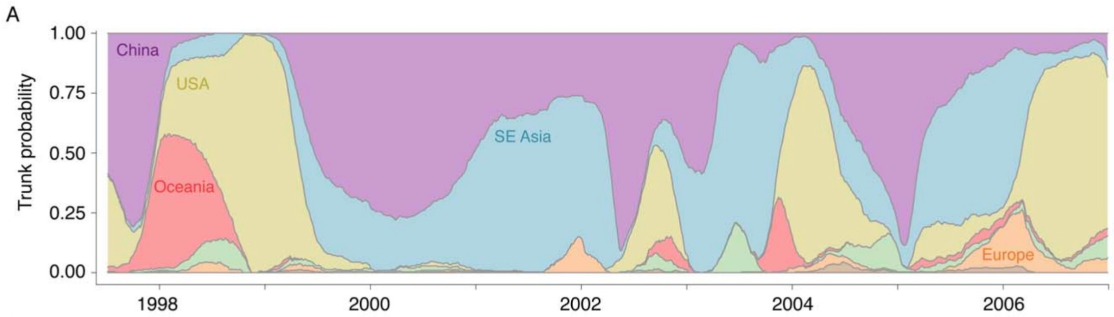




Lemey et al. 2009. PLoS Comp Biol

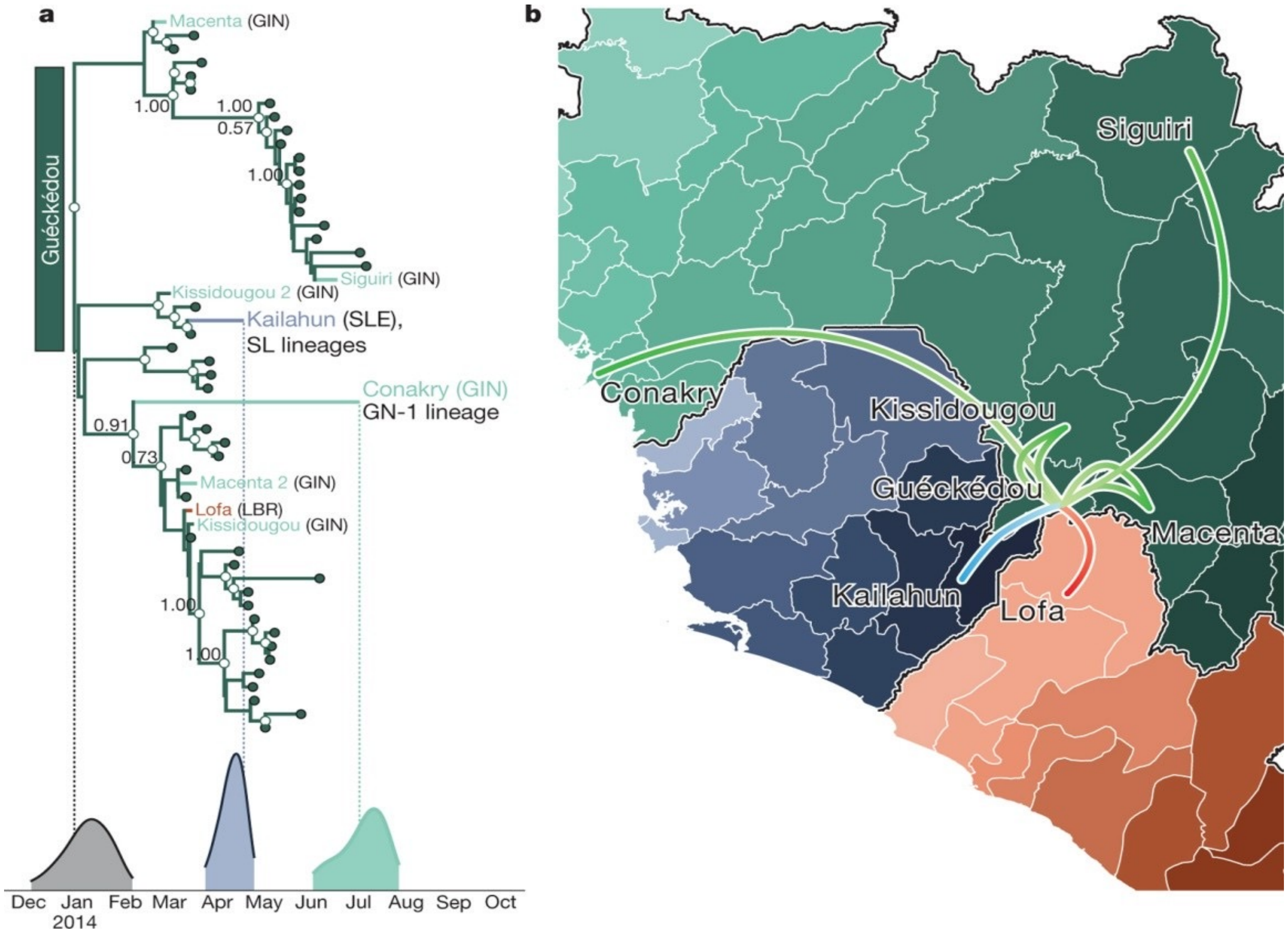
Global migration of Human Influenza A H3N2





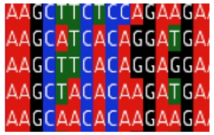
Bedford et al. 2010. PLoS Path

Early EBOV Dispersal in the 2014 Epidemic



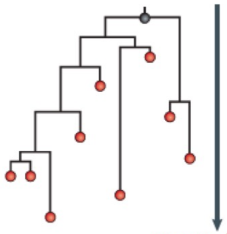
Dudas et al. 2017. Nature

Summary



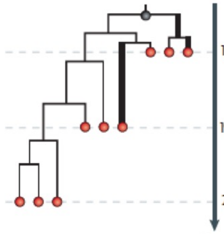
Pathogen Gene Sequences

Model of Nucleotide Substitution



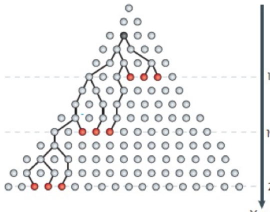
Evolutionary Tree

Molecular Clock Model



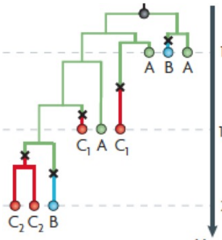
Dated Evolutionary Tree

Demographic Model



Population Dynamics

Translocation Model



Geographic Dispersal

Pybus & Rambaut. 2009. Nat rev Genet

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